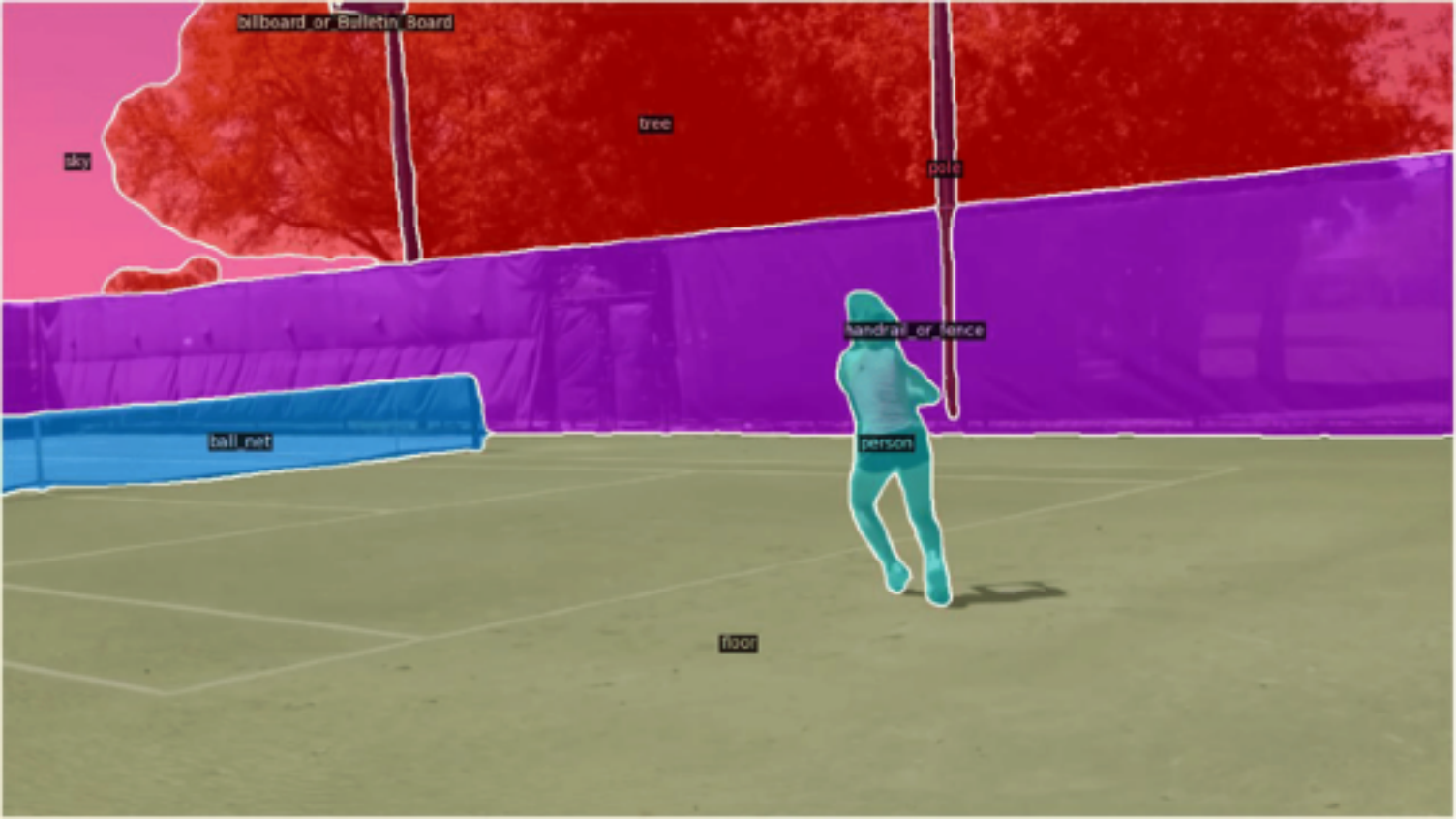
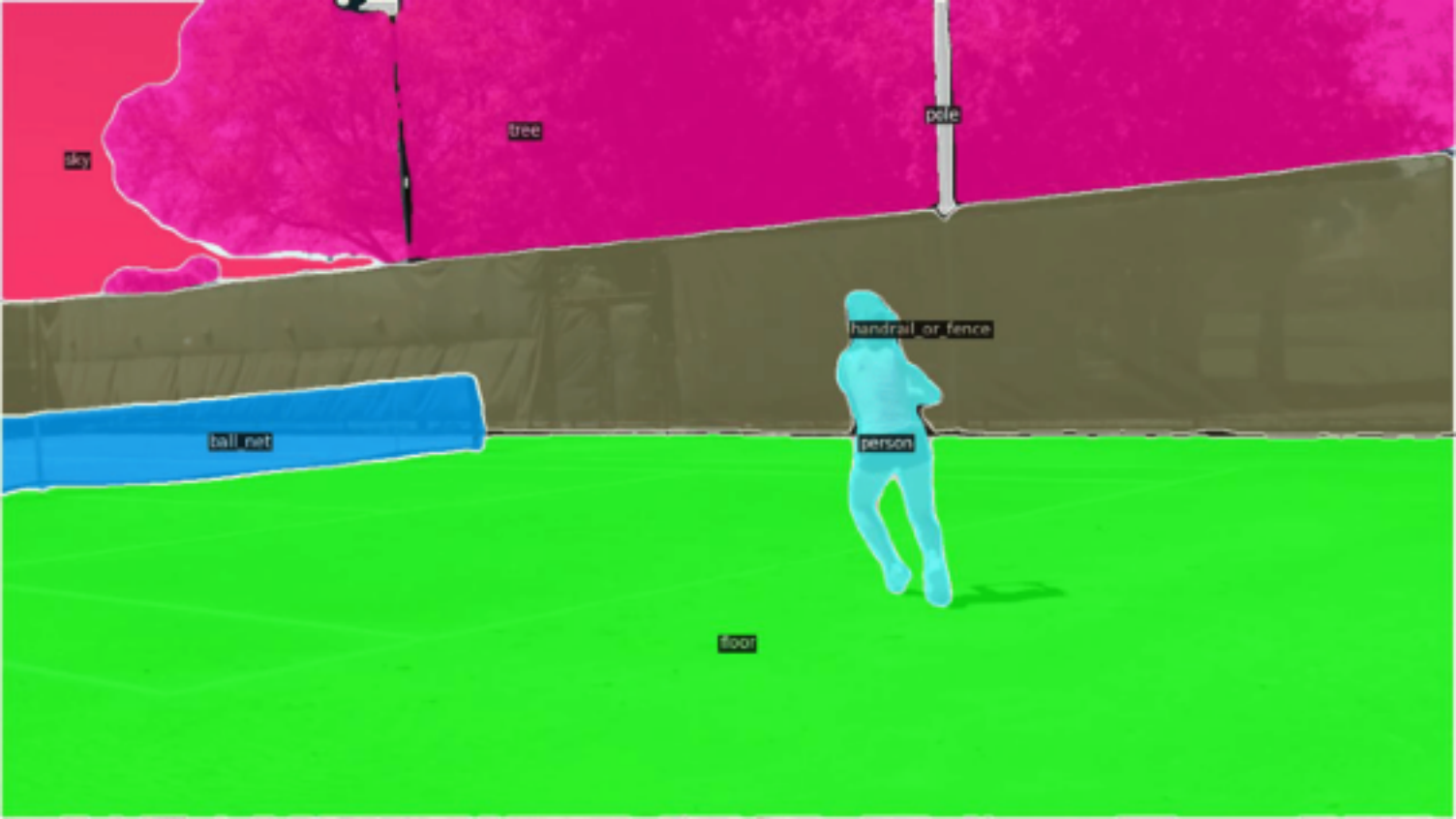
person 92%


billboard or Bulletin Board
tree
sky
pole
handrail or fence
ball net
person
floor


sky
tree
pole
handrail_or_fence
ball_net
person
floor

Video Instance Segmentation (VIS)

Video Semantic Segmentation (VSS)

Video Panoptic Segmentation (VPS)

Category names:

- **Thing: Person, *Tennis racket***
- **Stuff: Sky, Tree, Ground, Wall, …**

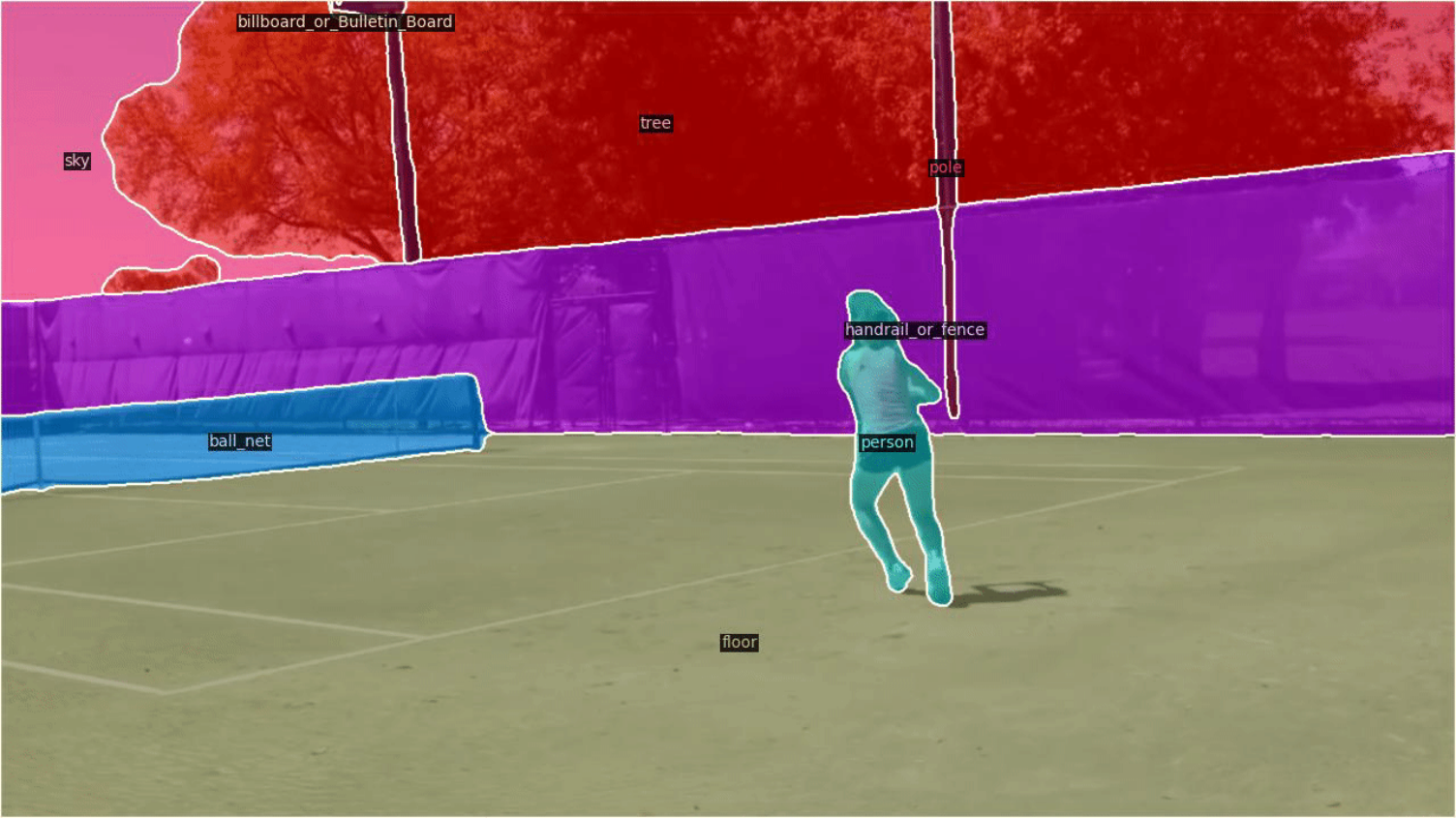
Category-guided Segmentation

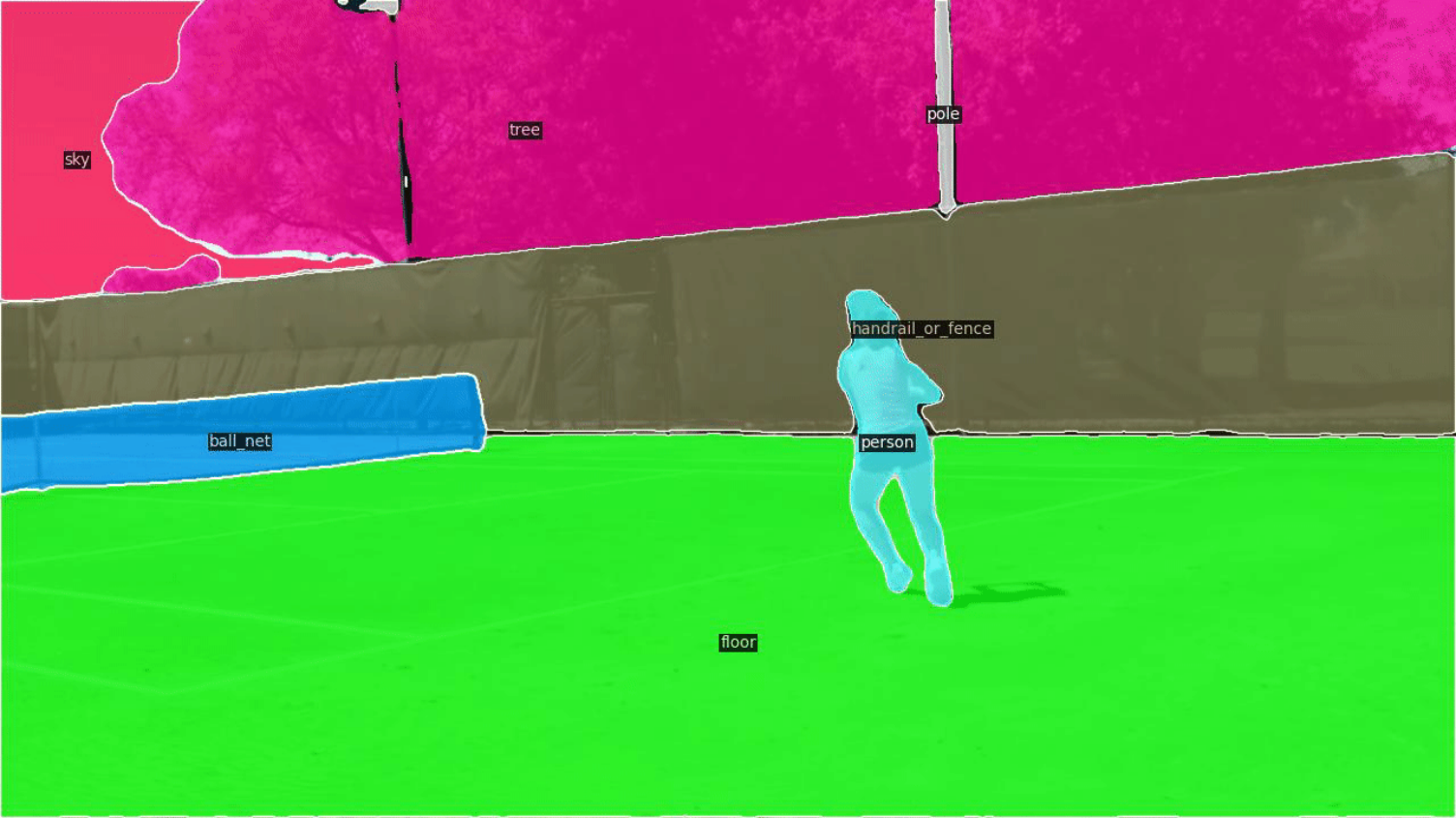
Better temporal consistency!!

UniVS: Video Segmentation Foundation Model

Minghan Li. **UniVS:** Unified Video Segmentation with Prompts (CVPR 2024)

person 92%


billboard_or_Bulletin_Board
tree
sky
pole
handrail_or_fence
ball_net
person
floor


sky
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Category-guided Segmentation

Video Instance Segmentation (VIS)

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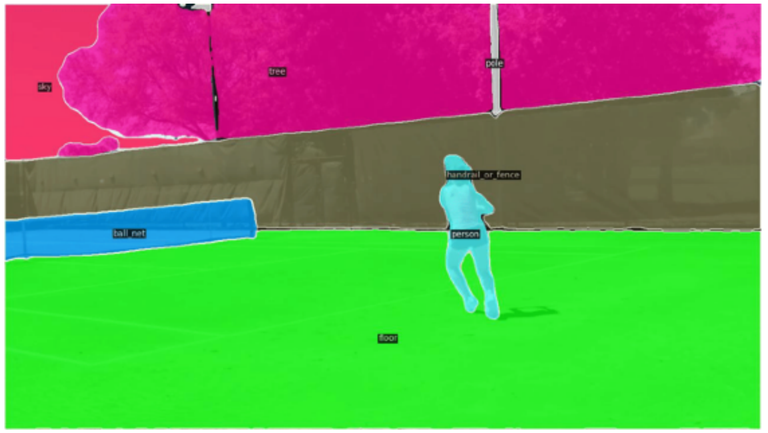


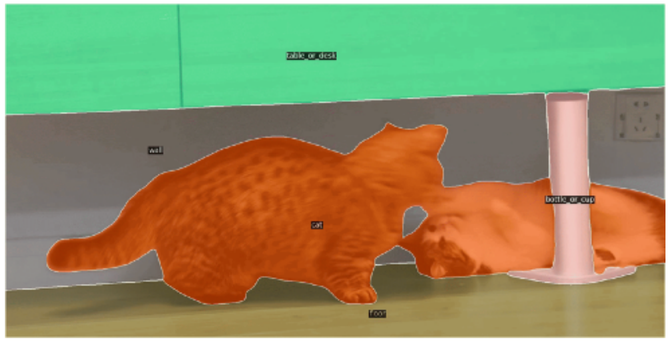
Video Panoptic Segmentation (VPS)

Better temporal consistency!!

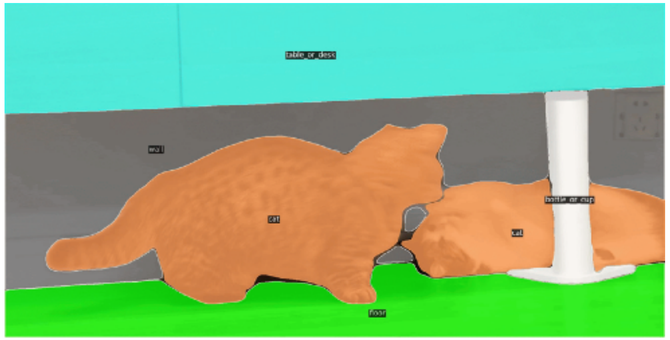
Minghan Li. **UniVS:** Unified Video Segmentation with Prompts (CVPR 2024)

UniVS: Video Segmentation Foundation Model

Video Instance Segmentation (VIS)



Video Semantic Segmentation (VSS)



Video Panoptic Segmentation (VPS)

Visual prompt

Text prompt

A cat quickly darting across the camera.

Video Object Segmentation (VOS)

Category-guided Segmentation

Prompt-guided Segmentation

Consistent understanding across different tasks

Minghan Li. **UniVS:** Unified Video Segmentation with Prompts (CVPR 2024)